

POISONING/TOXIC INJURIES

ALWAYS BE CAUTIOUS IN SUSPECTED POISONING TO AVOID RESCUER EXPOSURE. USE CBRN PROTOCOLS IN MULTIPLE CASUALTIES.

STEP 1-2-3 APPROACH – DON APPROPRIATE PPE BEFORE APPROACHING PATIENT/SCENE

STEP 1 ONE PERSON INCAPACITATED WITH NO OBVIOUS REASON	STEP 2 TWO PEOPLE INCAPACITATED WITH NO OBVIOUS REASON	STEP 3 ≥3 PEOPLE IN CLOSE PROXIMITY, INCAPACITATED WITH NO OBVIOUS REASON
APPROACH USING STANDARD PROTOCOLS	APPROACH WITH CAUTION USING STANDARD PROTOCOLS	USE CAUTION AND DECLARE POSSIBLE CBRN/EXPOSURE THREAT

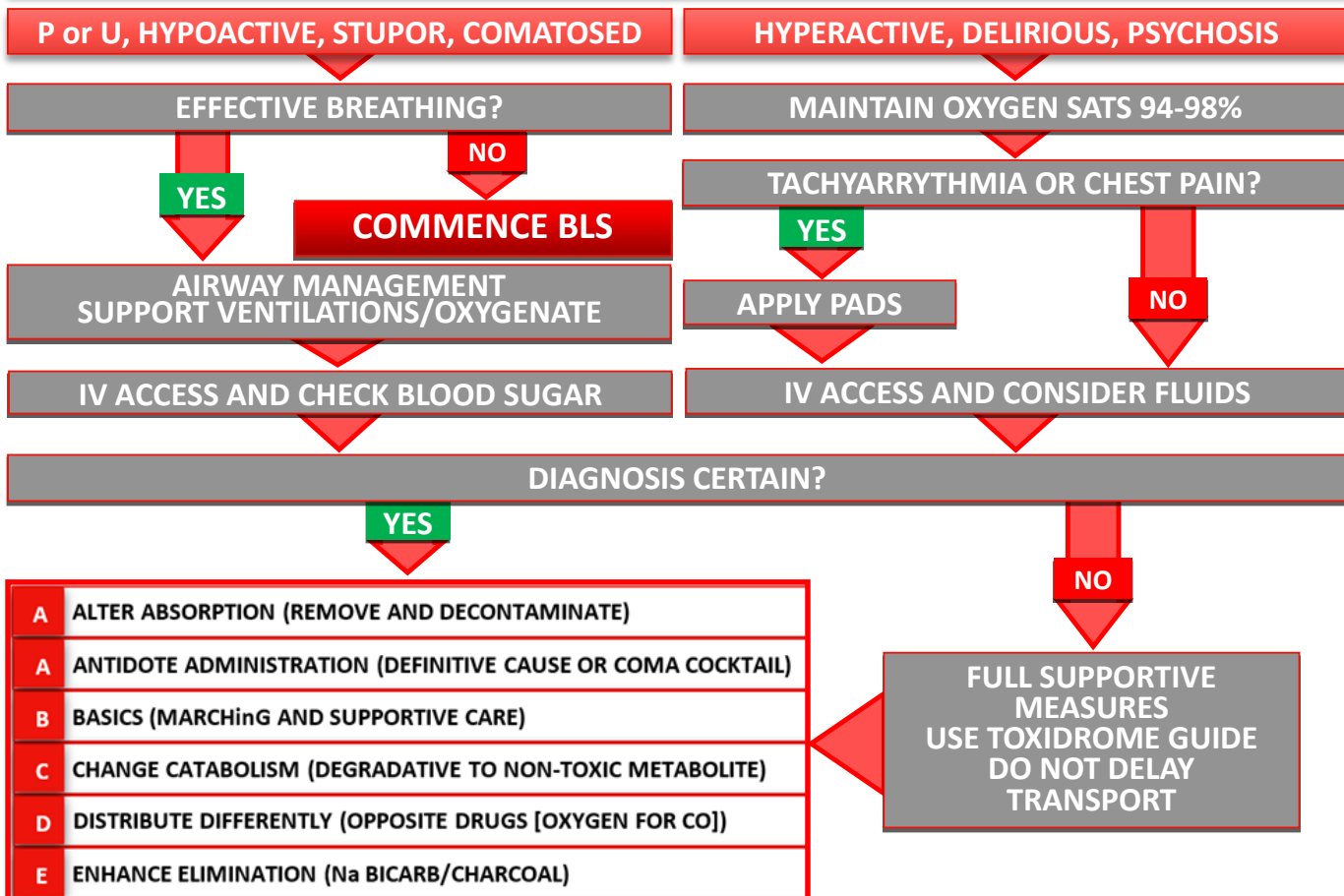
MANAGEMENT IN CBRN HOT ZONE (M²A²D ALGORITHM)

M	MASSIVE BLEED MASK / FACE (CRESS)	A	AIRWAY ANTIDOTES	D	DECONTAMINATE
---	-----------------------------------	---	------------------	---	---------------

	NERVE	CYANIDE	MORPHINE	ATROPINE	SEPSIS	HEAT STROKE
Conscious	Fitting	↓↓ / Fitting	↓	↑	Normal / ↓	Altered
Respiration	↑↑ / ↓	↑↑ / ↓↓	↓↓	N / ↑	↑↑	↑↑
Eyes	Pinpoint*	Normal / Large	Pinpoint	Large	Normal	Normal / Large
Secretions	↑↑	Normal	Normal	Dry	Normal / sputum	Normal
Skin	Sweaty	Pink / Blue	Normal / Blue	Flushed	Warm / Pale	Varied
Other potential features	Vomiting Incontinence Slow heart rate	Sudden onset			↑ pulse Fever (>38.3°C) Bio-syndrome	

* Pinpoint pupils may not be present immediately if skin absorption or eye protection worn.
Bio-syndromes include respiratory, cutaneous, lymphadenopathy, haemorrhagic, gastrointestinal & neurological.

MANAGEMENT IN PERMISSIVE ENVIRONMENT OR SINGLE SUSPECTED OVERDOSE PATIENT



TOXIDROMES							
	OPIOIDS	SEDATIVES	MUSCARINIC	CARBON MONOXIDE	SYMPATHO-MIMETIC	ANTICHOLIN-ERGIC	SEROTONIN SYNDROME
AIRW	DECREASED REFLEXES	DECREASED REFLEXES	INCREASED SECRETIONS	CLEAR	CLEAR	DRY MUCOUS MEMBRANES	CLEAR
RESP	BR ↓	BR ↓	BRONCHO-SPASM	BR ↑↑ SPO ₂ ↑↑	BR ↑	BR NORMAL / ↑	BR NORMAL / ↑
CIRC	HR ↓ BP ↓↓	HR ↓ BP NORMAL/↓	HR ↓↓ BP ↓	HR ↑ BP ↑/↓	HR ↑↑ BP ↑ CHEST PAIN	HR ↑ BP ↑	HR ↑ BP ↑
HEAD	P/U	P/U	P/U or FITTING	IRRITABLE / DROWSY	HYPERACTIVE	DELIRIUM	AGITATED THEN P/U
PUPILS	PINPOINT	NORMAL	PINPOINT	NORMAL	DILATED	DILATED	DILATED
HEAT	NORMAL	NORMAL	NORMAL	NORMAL	HIGH	HIGH	RAISED
SKIN	NORMAL/ CYANOSED	NORMAL/ CYANOSED	CLAMMY++	FLUSHED	CLAMMY ++	DRY / FLUSHED	CLAMMY
OTHER	IV MARKS?	HYPO- REFLEXIA	URINATED/ VOMITED	HEADACHE++ NAUSEA VERTIGO	TACHY- ARRHYTHMIA	HYPOACTIVE BOWEL SOUNDS	HYPER- REFLEXIA

TOXIDROME TREATMENTS - REFER TO MEDICATION GUIDE/BNF

SYMPATHOMIMETICS

COCAINE, AMPHETAMINE, EPHEDRINE, PHENCYCLIDINE, MEPHEDRONE, CATHINONE

- CONTROL AGITATION = SEDATE?
- CHECK ECG FOR POSSIBLE MI
- BENZODIAZEPINES (DIAZEPAM 10mg)
- HYPONATRAEMIA TREATMENT

ANTICHOLINERGICS

ATROPINE, TRICYCLIC ANTIDEPRESSANT (TCA), ANTIHISTAMINE

- CONTROL DELIRIUM = SEDATE?
- PHYSOSTIGMINE 2mg (SLOW IV) FOR ATROPINE OVERDOSE
- SODIUM BICARBONATE 8.4% 50ml FOR TCA OVERDOSE

MUSCARINICS

NERVE AGENTS, ORGANOPHOSPHATES, CARBOMATES, CHOLINOMIMETICS

- COMBOPEN AUTOINJECTOR
- ATROPINE (1-3mg EVERY 5 MINS UNTIL BRONCHORRHEA STOPS)
- PRALIDOXIME (UP TO 2g)
- BENZODIAZEPINES FOR FITTING

OPIOIDS

HEROIN, MORPHINE, FENTANYL, CODEINE, METHADONE

- VENTILATE AND OXYGENATE
- NALOXONE (400mcg SLOW IV or 800mcg IM) UNTIL GOOD SPONTANEOUS RESP FUNCTION ACHIEVED AND SUITABLE RESPONSE LEVELS

SEDATIVES

BENZODIAZEPINES, BARBITUATES, SLEEP MEDS, ANAESTHETICS, ALCOHOL

- VENTILATE AND OXYGENATE
- FLUMAZENIL 200mcg (SLOW IV) UP TO 3mg FOR BENZO OVERDOSE
- IV INTRALIPID EMULSION FOR LOCAL ANAESTHETIC TOXICITY

ASPHYXIANTS

CYANIDE, CARBON MONOXIDE, HYDROGEN SULPHIDE

- REMOVE FROM SOURCE
- VENTILATE AND OXYGENATE
- AMYL NITRATE/SODIUM THIO-SULPHATE/HYDROXYCOBALAMIN FOR CYANIDE POISONING

UNKNOWN POISONING SYMPTOMS TREATMENT

FITTING

BENZODIAZEPINES FIRST LINE
- DIAZEPAM 10mg IV or MIDAZOLAM 10mg IM (5mg IV)
OXYGEN 15L/min

MEMBRANE STABILISATION

MAGNESIUM 2g IV/IO OVER 20 MINUTES
DILUTE 1ml MAGNESIUM WITH 9ml SALINE/5% GLUCOSE

URINE ALKILINISATION

SODIUM BICARBONATE 8.4% 50ml
OR
SODIUM BICARBONATE 1.6% 500ml

TACHYARRHYTHMIA

ANY SHOCK, CHEST PAIN, SYNCOPE OR HEART FAILURE NEEDS URGENT CARDIOVERSION

- FOLLOW TACHYARRHYTHMIA ALS
- CONSIDER MEMBRANE STABILISATION
- TOXICITY INDUCED TACHYARRHYTHMIA MAY NOT RESPOND TO DEFIBRILLATION

BRADYARRHYTHMIA

ANY SHOCK, CHEST PAIN, SYNCOPE OR HEART FAILURE NEEDS URGENT PACING

- FOLLOW BRADYARRHYTHMIA ALS
- CONSIDER ATROPINE (0.5mg-2mg EVERY 10-15mins)
- CONSIDER TRANSCUTANEOUS PACING

METHEMOGLOBINAEMIA

LOCAL ANAESTHETIC, NITRATES, SOME ABX, METOCLOPRAMIDE, ANILINE, DAPSONE

- OXYGEN 10-15L/min
- METHYLENE BLUE 1-2ml/kg